

CHARLES (CHUCK) FORSYTH

Research Computing and AI Executive | HPC | Research Cloud | Navy Veteran

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SUMMARY

Research computing and AI executive with 30+ years in technology, from Navy radar to high-performance computing to generative AI. As Director of Research Computing at UC Riverside, I lead a matrix team of about 20 across ITS and the colleges and run the cloud, cluster, storage, security and AI services behind about 1,000 researchers in 22 colleges and schools, and I still build the platforms myself, now with a team of AI agents. I led UCR's first-of-its-kind research partnership with Google Cloud, served as Co-PI on a \$498K NSF CC* research storage award, and chaired the UC Research IT Committee across every UC campus. I have been building with AI since the GPT-2 era, running GPU jobs for it on the National Research Platform in 2020, and shipped an LLM assistant for HPC users in 2023. The number I manage to is time to science.

CAREER HIGHLIGHTS

- **Google Cloud partnership:** led Ursa Major, announced by Google in 2023 as a first-of-its-kind research subscription; 170+ research projects under one governed, budgeted cloud.
- **NSF award:** Co-PI on CC* award 2346636 (\$498,553), which delivered 2.09 PB of research storage on the 100 Gbps Science DMZ, 422 TB shared with the National Research Platform.
- **AI for research:** built UCR's governed AI stack (MCP servers, AI Gateway, elastic GPU cluster) and ship it with AI agents: 240+ releases across ten tools in five weeks.
- **Campus and systemwide leadership:** founded and lead UCR Research IT Leaders; Research Advisory Board and Academic Senate research committee; Chair of the UC Research IT Committee (2022-2024), uniting research IT leaders from every UC campus.

PROFESSIONAL EXPERIENCE

Director of Research Computing

Mar 2023 - Present

University of California, Riverside, Riverside, CA

Lead research computing strategy, services and AI for about 1,000 researchers in 22 colleges and schools through a matrix team of about 20: direct reports and student staff, the ITS systems team (dotted line for research) and research IT staff in colleges, centers and units across campus.

- Run Ursa Major, the Google Cloud research partnership: per-project budgets, automated spend throttles, security baselines and FinOps across 170+ projects.
- Wrote NSF CC* proposal 2346636 and its budget (Co-PI); designed CephRDS (2.09 PB usable) and its cost recovery model; six science driver labs migrated, 422 TB federated to the National Research Platform.
- Built the AI-for-research stack: MCP servers with per-user Google sign-in, roles and a full audit trail; the UCR AI Gateway; and an elastic Slurm cluster on Google Cloud (CPU, high-memory, NVIDIA L4 GPU) with zero compute cost when idle.
- Built Nexus, the system of record Research Computing runs on: 943 researchers, 431 org units, 176 cloud projects, 12,000+ links.
- Data security plans and secure enclaves for P3/P4 and CMMC research with the ISO.
- Founded and lead UCR Research IT Leaders, which coordinates research IT across campus; serve on the Research Advisory Board and the Academic Senate research committee.
- Official UCR PI status (2024); grant strategy; ship software with a team of AI agents under written specs, test gates and human approval.
- Rebuilt the Research Computing website and service catalog (live Oct 2026).
- Built nrp-mcp (AI-assisted jobs on the National Research Platform, pilot) and Skywalker, the audit engine for UCR's research cloud.

Associate Director of Research Computing

Aug 2019 - Mar 2023

University of California, Riverside, Riverside, CA

- Developed research computing grant strategy and managed budgets for research technology initiatives within Information Technology Solutions (ITS).
- Lead systems architect for research computation, advanced statistical computing and network interoperability with the campus Science DMZ.
- Served as proxy for the CIO and CTO in academic senate technology discussions.
- Directed a team of professionals in HPC, data and cyberinfrastructure research and technology projects.
- Supervised and mentored technical staff; set fiscal, HR and resource allocation policy for the unit.
- Coordinated research technology efforts within ITS with BCOE, HPCC and library partners.
- Onboarded UCR researchers to the Pacific Research Platform's Nautilus GPU cluster: Kubernetes job templates, TensorFlow and PyTorch GPU pods (2019), GPT-2 GPU jobs on NVIDIA RTX 8000 (2020) and UCR Nautilus cluster tools.
- Published Google Colab notebooks for researchers: action recognition, object detection and image enhancement (2020), Stable Diffusion and Disco Diffusion (2022).
- Ran a Terraform infrastructure-as-code workshop for AWS and Google Cloud (2020).

Chair, UC Research IT Committee

Jul 2022 - Jan 2024

University of California (systemwide), Remote

Vice Chair, Jun 2021 - Jul 2022.

- Led the UC Research IT Committee, liaising with the UC IT Leadership Council (ITLC) and the UC Office of the President CIO.
- United research IT leaders from every UC campus around shared priorities; set and met annual goals and advocated systemwide for UC researchers' computing needs.

Lead Linux & HPC Systems Administrator / Consultant

Jun 2017 - Aug 2019

UC Riverside, High Performance Computing Center (HPCC), Riverside, CA

- Rebuilt the HPCC campus cluster and the server room it runs in, including new power and cooling.
- Moved cluster configuration management to Ansible and set up PXE bare-metal provisioning.
- Architected, installed, administered and supported UCR's campus HPC systems.
- Built the HPCC Cloud Service (2017-2018): lab-private HPC clusters in AWS on demand, created from the HPCC cluster with `hpcc_cloud`, a `cfnCluster` wrapper I wrote (GPU, high-memory and autoscaling nodes, per-minute billing, budget alerts); wrote its manuals and maintained its software modules.
- Maintained the cluster's software environment (130 commits to the HPCC module tree, 2017-2019) and the HPCC website (94 commits).
- XSEDE Campus Champion: connected UCR researchers and students to NSF supercomputing centers and national laboratories.
- Taught 19 HPCC workshops for UCR researchers and students (Jul 2017 - Jun 2019; sole instructor on 10): Linux and cluster basics, Python, software installation, and an HPCC + AWS cloud cluster service; wrote the four training decks.
- UCR representative to the UC Research IT Committee and to the Campus Research Computing Consortium (CaRCC).

Lead Linux & HPC Systems Administrator / Research Consultant

Jan 2012 - Jun 2017

Corning Incorporated, Corning, NY

Lead HPC architect and administrator for Corning's research, development and engineering computing.

- Linux clusters with thousands of cores; Lustre, InfiniBand, SAN and NAS storage.
- 60+ network-licensed science codes (Ansys, COMSOL, Abaqus; Intel MPI, OpenMPI; MKL, FFTW) under FlexLM, RLM, LM-X and MathLM.
- 400+ server, storage and network support contracts; 150 kVA+ of data center power and cooling.
- Provisioning and configuration (iLO, Cobbler, iPXE, Kickstart); monitoring (Nagios, ELK); virtualization (VMware, XenServer, oVirt KVM); OpenStack and AWS.
- Led internal and external projects; customer training and documentation.

Owner and Research System Administrator

Jun 2003 - Jan 2012

CnL Networks, Whitney Point, NY

- Founded and ran a systems administration and IT outsourcing business: billing, accounting, infrastructure and support systems.
- Research computing support, Linux and Windows servers, email and web hosting, networks and procurement consulting for individuals and small businesses.

Technical Account Manager

Apr 2008 - Aug 2009

STCR, Endicott, NY

Linux System Administrator

Mar 1998 - Jun 2003

Hancock Telephone Company, Hancock, NY

- Web, email, NAS, DNS, RADIUS and file servers, firewalls, intrusion detection and packet inspection for an internet provider; customer support for connectivity and hosting.

Hardware Test Engineer

Aug 1997 - Mar 1998

IBM, Endicott, NY

- Hardware test, troubleshooting and root cause analysis; documented performance trends for production.

Technical Support

Mar 1997 - Aug 1997

AOL, Jacksonville, FL

- Dial-up and DSL support; led internal training sessions.

Advanced Electronics Technician

Dec 1994 - Mar 1997

United States Navy, USS Lake Champlain (CG-57), Norfolk, VA

- System- to component-level troubleshooting and maintenance of radar and communications systems, including IFF (Identify Friend or Foe), aboard an Aegis cruiser.
- Trained operators; managed IFF maintenance records and testing.

GRANTS

- National Science Foundation, CC* Data Storage: Foundational Campus Research Storage for Digital Transformation, award 2346636, \$498,553, Mar 2024 - Feb 2026. Co-PI. Wrote the full proposal, budget and justification, and the PI and Co-PI documents under the NSF PAPPG, in one month. Delivered CephRDS: 2.09 PB usable (3.2 PB raw), 20 nodes on the 100 Gbps Science DMZ, 422 TB federated to the National Research Platform.

AI WORK TIMELINE

- **2019:** TensorFlow and PyTorch GPU pods for UCR researchers on the National Research Platform.
- **2020:** GPT-2 GPU jobs (NVIDIA RTX 8000) on Nautilus; deep learning notebooks for researchers.
- **2022:** Stable Diffusion and Disco Diffusion notebooks for researchers, before ChatGPT launched.
- **2023:** Ursa Major Ask, a GPT-3.5 assistant that writes and runs research code and Slurm scripts.
- **2024:** uma, an LLM in the Linux shell for HPC users.
- **2025:** Gemini multimodal video analysis, multi-agent simulations, generative media tools, deep-research.
- **2026:** MCP servers for HPC, research systems and NRP; UCR AI Gateway; Directed Agentic Engineering.

AI AND RESEARCH PLATFORMS BUILT

- **ursa-bifrost (2026):** MCP server and CLI in one Go binary that lets Claude, Gemini, Codex and other AI assistants work UCR's Slurm cluster through typed, audited, read-mostly tools; submit and cancel are opt-in with two-step approval and spend caps; per-user Google sign-in; hosted on Cloud Run. MIT licensed. github.com/UCR-Research-Computing/ursa-bifrost
- **UCR AI Gateway (2026):** One governed OpenAI- and Anthropic-compatible endpoint to Gemini, Claude, open-weight and National Research Platform models, with per-lab budgets, per-key spend limits, rate limits, a full spend log and an admin MCP server.
- **Nexus (2026):** The system of record for Research Computing's relationships: a Postgres and pgvector knowledge graph of researchers, labs, cloud projects, grants, tasks and interactions, run through a CLI and an OAuth MCP server with 41 role-gated tools.

- **Ursa Major HPC cluster on Google Cloud (2026):** Elastic Slurm cluster built with Cluster Toolkit: compute created on demand and deleted when idle; CPU, high-memory and NVIDIA L4 GPU partitions; per-lab accounts and budget guardrails.
- **deep-research (2025-):** CLI and web workstation for Gemini Deep Research that produces cited reports and checks their claims as real jobs on the HPC cluster. MIT licensed. github.com/charles-forsyth/deep-research
- **Ultra AI Workstation (2026):** Keyboard-driven AI operator desk for email, calendar, tickets, Slack, the Nexus ledger and the cluster, with AI drafts under two-step approval. MIT licensed. github.com/charles-forsyth/ultra-workstation
- **Skywalker (2025-):** Audit engine that scans hundreds of Google Cloud projects for security risk, idle and orphaned resources and AI spend; also a read-only MCP server. github.com/charles-forsyth/Skywalker
- **ursa-dsp (2026):** Data Security Plan case tracker and assembler built with UCR's Information Security Office; people make every decision.
- **nrp-mcp (2026):** Local MCP server for AI-assisted jobs on the NRP Nautilus GPU cluster, with NRP policy built in and single-use approval. Pilot.
- **UCR HPC Job Gallery (2026):** 49 tested, standardized Slurm workflows across seven colleges and schools (including AlphaFold, from bioengineering to social science), with README, index and HTML gallery generated from the scripts. github.com/charles-forsyth/UCR-HPC-Job-Gallery
- **squad-manager (2026):** Directed Agentic Engineering: orchestrates squads of specialized AI agents (architect, test writer, DevOps, developer, security gatekeeper, user acceptance) to design, build, test and release software. github.com/charles-forsyth/squad-manager
- **Generative AI media tools (2025):** CLIs for Google Veo 3.1 video, Lyria music and Gemini TTS, and a Gemini multimodal video analyzer for research footage. github.com/UCR-Research-Computing/gemini-video-analyzer
- **TPU and accelerator demos (2026):** JAX/Flax demos on Google Cloud TPU v5e for science disciplines and a CUDA-to-TPU porting demo. github.com/charles-forsyth/TPU-Demos
- **Browser AI demos (2025):** Live webcam hand tracking (MediaPipe), reinforcement-learning game agents, an HPC workflow simulator and a public voice AI agent. lordivxx.github.io
- **Ursa Major Ask (2023) and uma (2024):** Early generative AI assistant that let researchers ask about the research cloud and generate and run code and Slurm scripts, followed by uma, an LLM in the Linux shell. github.com/UCR-Research-Computing/uma

PUBLICATIONS

- "One Day on the Research Stack," Field Notes, Oct 2026. charles-forsyth.github.io/papers/
- The AI + HPC Field Manual, 15 chapters, 2026. charles-forsyth.github.io/handbook/

TEACHING AND TRAINING

- HPCC workshops, UCR High Performance Computing Center, 19 sessions, Jul 2017 - Jun 2019 (sole instructor on 10; others with Jordan Hayes and Thomas Girke), Genomics Building Auditorium: HPCC Cluster Intro (2017-2018); Linux Basics and HPCC Cluster Introduction (2018-2019); Basic Linux, Cluster Intro and Software Installation (Sep 2018); Introduction to Python (Oct 2018); Research Computing in UCR's HPC Center (Oct 2018). Authored decks: HPCC Cluster Intro (43 slides), Linux Basics and HPCC Cluster Intro (53), Introduction to Python (30).
- HPCC Cloud Service presentation: HPC clusters in AWS on demand, HPCC AWS Cluster deck (36 slides), Apr 2018. Workshop list and decks: hpcc.ucr.edu/events/events/
- Connecting to XSEDE resources (Stampede2); Windows applications on Linux clusters (2017-2019)
- Terraform infrastructure-as-code workshop, AWS and Google Cloud (2020)
- National Research Platform (Nautilus) researcher guide, 8 articles (UCR KB023-KB030, 2026)
- Training decks and recordings archived at charles-forsyth.github.io/handbook/

PROFESSIONAL SERVICE AND LEADERSHIP

- Chair, UC Research IT Committee (Jul 2022 - Jan 2024)
- Vice Chair, UC Research IT Committee (Jun 2021 - Jul 2022)

- Founder and Chair, UCR Research IT Leaders (RITL), coordinating research IT across campus (Present)
- Member, UCR Research Advisory Board (Present)
- Member, UCR Academic Senate research committee (Present)
- Official UCR Principal Investigator status (2024 - Present)
- UCR Campus Champion, NSF XSEDE and ACCESS (Jun 2017 - Present)
- NSF grant peer reviewer (May 2020 - May 2021)
- Co-Chair, UCR Visualization Laboratory Steering Committee (Aug 2019 - Present)
- UCR representative, Campus Research Computing Consortium (CaRCC) (Jun 2017 - Present)
- UCR representative, Coalition for Academic Scientific Computation (CASC) (Aug 2019 - Present)
- UCR representative, Advanced Cyberinfrastructure Research and Education Facilitators (ACI-REF) (Aug 2019 - Present)
- Research collaborator, Pacific Research Platform (Aug 2019 - Present)

EDUCATION

- **B.S., Information Technology**, Purdue University Global (formerly Kaplan University), 2015. GPA 3.89. President's List (four terms, 2013-2014); Golden Key International, Iota Sigma Tau and Alpha Beta Kappa honor societies.
- **Networking and Technical Support Specialist**, Ridley-Lowell Business & Technical Institute, 2011. GPA 4.0. Highest GPA Achievement Award; Dean's List; High Honor Roll.
- **Advanced Electronics Technology**, United States Navy, 1997. AC/DC and solid state circuits, digital microprocessors, radio and radar, communications systems and troubleshooting.

CERTIFICATIONS

- PBS Professional Administrator (Altair)
- Mellanox Certified InfiniBand Professional, Level 1
- EMC Isilon Administration
- Kubernetes: Cloud Native Ecosystem; Learning Kubernetes (2020)
- Project Management Foundations (PMI)
- UCR Organizational Excellence: Bronze Certificate and Collaborative Leadership (2018)
- Professional development in leadership, project management and remote work, PMI and LinkedIn Learning (2018-2020)

TECHNICAL EXPERTISE

- **AI:** MCP servers, agentic engineering, Gemini, Claude and open-weight models, Vertex AI, RAG and pgvector
- **HPC:** Slurm, PBS Professional, MPI, InfiniBand, Lustre, GPFS/Spectrum Scale, Ceph, Isilon, DDN
- **Cloud:** Google Cloud (Cluster Toolkit, Cloud Run, Cloud SQL, GKE, IAM, FinOps), AWS, OpenStack, Kubernetes
- **Systems:** Linux (CentOS/RHEL 4 onward), Windows Server, VMware, XenServer, KVM, Nagios, ELK
- **Languages:** Python, Go, Bash, C++, C#, PowerShell, SQL